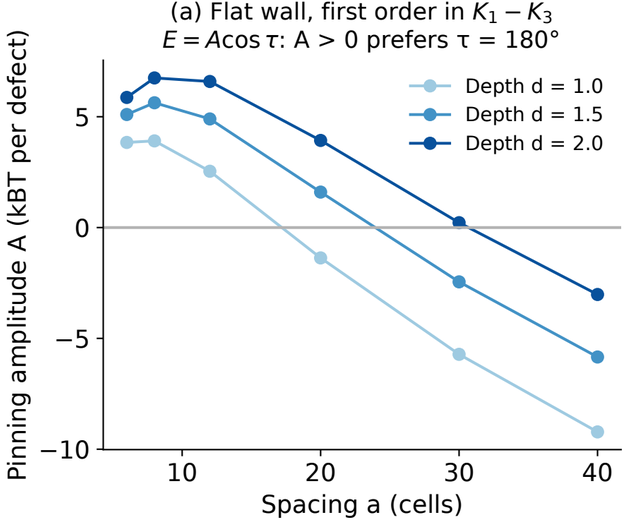
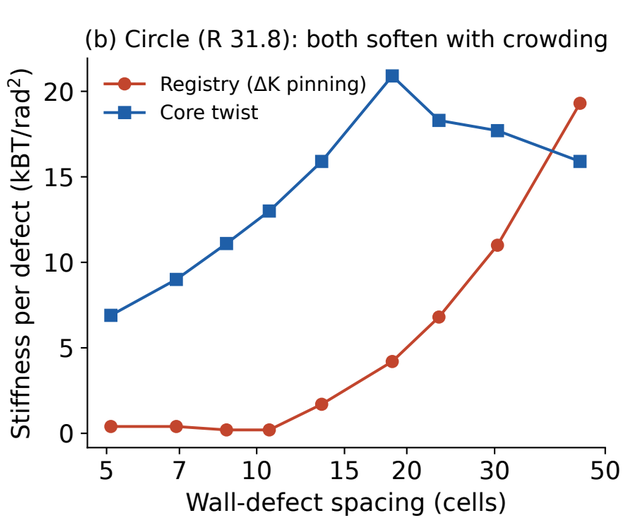
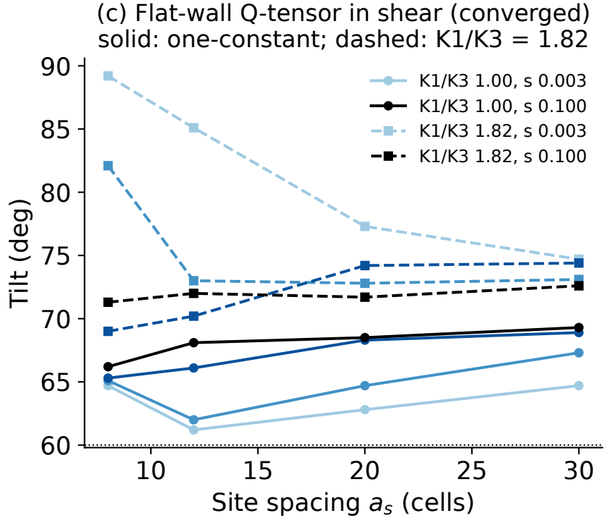
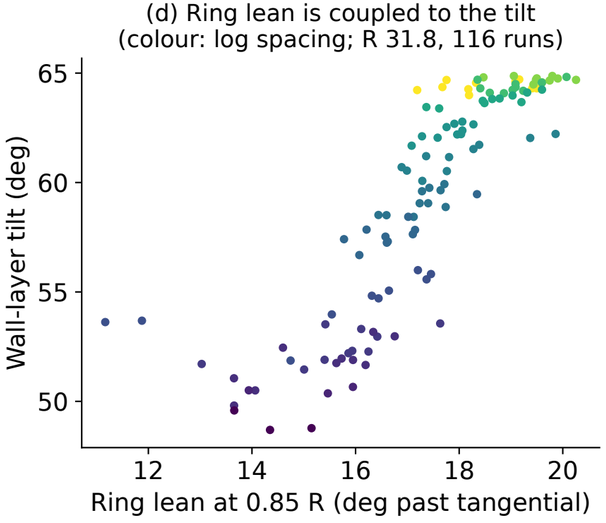

(a) Flat wall, first order in $K_1 - K_3$
$E = A\cos\tau$: A > 0 prefers τ = 180°
Depth d = 1.0
Depth d = 1.5
Depth d = 2.0
Pinning amplitude A (kBT per defect)
Spacing a (cells)


(b) Circle (R 31.8): both soften with crowding
Registry (ΔK pinning)
Core twist
Stiffness per defect (kBT/rad²)
Wall-defect spacing (cells)


(c) Flat-wall Q-tensor in shear (converged)
solid: one-constant; dashed: K1/K3 = 1.82
K1/K3 1.00, s 0.003
K1/K3 1.00, s 0.100
K1/K3 1.82, s 0.003
K1/K3 1.82, s 0.100
Tilt (deg)
Site spacing a_s (cells)


(d) Ring lean is coupled to the tilt
(colour: log spacing; R 31.8, 116 runs)
Wall-layer tilt (deg)
Ring lean at 0.85 R (deg past tangential)